

Mohammad Sharif

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Greater Chicago Area

Strong letters of recommendation available upon request.

EDUCATION

Illinois Institute of Technology

Chicago, IL

Bachelor of Science in Mechanical Engineering | GPA: 3.6

August 2024 - May 2027

- Relevant Coursework: Fluid Mechanics, Thermodynamics, Computational Mechanics, Advanced Mechanics of Solids, Material Science, Electromagnetism, Differential Equations, Multivariable Calculus
- Honors: Dean's List, Heald Scholar, STEM Scholar, Emerging Campus Leader Scholar

EXPERIENCE

Project Engineering Co-Op, USG – Baltimore, MD

January 2026 - Present

- Owned the scope, planning, and execution of a \$340,000 equipment upgrade project focused on future-proofing a production department for long-term operational reliability
- Coordinated contractors, project timelines, safety compliance, procurement, and communication with plant leadership throughout all project phases
- Supported onboarding and training of incoming engineering co-ops by assisting with project coordination, plant procedures, and day-to-day engineering operations

Biomedical Engineering Intern, Project C.U.R.E. – Woodridge, IL

August 2025 - December 2025

- Ensured quality assurance of 100+ pieces of biomedical equipment from external medical facilities through inspection, testing, and repair using standard work instructions
- Contributed to 3 international medical shipments, including 60+ ICU beds, 15+ gurneys, 10+ home beds, infusion pumps, oxygen concentrators, and electrosurgical units to countries in need

Student Researcher, Illinois Institute of Technology – Chicago, IL

July 2025 - January 2026

- Designed a cost-effective produce slicer with simplified parts, developing CAD models and technical drawings in SolidWorks, achieving a 40% weight reduction
- Field-tested and prototyped 8+ iterations to optimize blade types and performance across multiple metrics
- Organized technical reports in Overleaf summarizing research and design outcomes for patent submission

PROJECTS

Bio-Inspired Quadruped Robot

October 2025 - December 2025

- Designed and assembled frog-inspired quadruped robot in Inventor with 4 servo motors controlling all gaits
- Achieved top 15% performance in class with .4 body lengths per second locomotion speed
- Optimized Arduino Uno code by determining servo max speeds and optimal torque application for efficient movement
- Created circuit integration connecting leg servos to Arduino controller and documented design in technical report

EV Charging Robotic Arm

September 2025 - November 2025

- Designed 4-bar linkage mechanism to autonomously reach EV charging port at designated distance
- Developed MATLAB optimization code to determine optimal ground locations, bar lengths, and joint angles for linkage geometry
- Fabricated mechanism through laser cutting and 3D printing, integrating Arduino-controlled servo with IR receiver for signal-based actuation and retraction
- Produced technical documentation in Overleaf detailing design process, calculations, and performance analysis

iGEM Bee Husbandry Design Engineer & Treasurer

August 2025 - December 2025

- Collaborated on the design of a bee monitoring enclosure for plasmid data collection
- Designed enclosure in SOLIDWORKS and supported fabrication via 3D printing, laser cutting, and CNC
- Served as project treasurer, managing budget tracking, and coordination with financials to report sending and ensure timely reimbursements

SKILLS

- Design & Simulation: SOLIDWORKS, AutoCAD, Inventor, DraftCAD, Orca Slicer
- Programming & Tools: Python, MATLAB, Arduino, LaTeX, Excel
- Languages: English, Urdu, Spanish